Name: Palak Singh **Roll No: 202403123** **Class: TYCS-B**

Practical 12

Checkbox Identification and Validation

Aim

To identify checkboxes on a web page, count checked and unchecked checkboxes, perform dynamic selection/deselection, and validate the checkbox states using Selenium WebDriver.

Software/Tools Required

- GitHub Codespaces
- Java
- Maven
- Selenium WebDriver
- Chrome browser
- JUnit

Step 1: Open the GitHub Repository

Open the practical repository provided by the teacher:

https://github.com/Conceicaodsouza/STQA-Practical-12-CheckboxValidation

Click – Code >> Codespace >> Create codespace on main

Step 2: Check Java

Open the Codespace terminal and type:

java -version

Java version information should be displayed.

Step 3: Check Maven

Type:

mvn -version

Maven version information should be displayed.

Step 4: Start the Local Web Server

In the terminal, type:

jwebserver -p 8000 -d /workspaces/STQA-Practical-12-CheckboxValidation/src/test/resources

You should see that the server is running on port **8000**.

Do not close this terminal.

Step 5: Forward Port 8000

In GitHub Codespaces:

1. Open the **PORTS** tab.
2. Click **Forward a Port**.
3. Enter:

8000

4. Open the forwarded port in the browser.

The checkbox web page should appear.

Step 6: Observe the Web Page

The page contains **6 checkboxes**:

1. Java – Checked
2. Python – Unchecked
3. Software Testing – Checked
4. Database Management – Unchecked
5. Web Development – Unchecked
6. Computer Networks – Unchecked

Therefore:

Total checkboxes : 6
Checked checkboxes : 2
Unchecked checkboxes: 4

Step 7: Check Chrome and Install Required Dependencies

First check whether Google Chrome is already installed:

google-chrome --version
If Chrome version is displayed

For example:

Google Chrome 154.0.8037.57
Chrome is available. **Continue to Step 8.**

If you get:

google-chrome: command not found
Selenium may still download its own Chrome automatically. First check whether Selenium has downloaded Chrome:

ls ~/.cache/selenium/chrome/linux64/
If a version number such as:

154.0.8037.92
is displayed, Chrome has been downloaded by Selenium.

Now install the required Chrome libraries:

sudo apt update

Then:

sudo apt install -y fonts-liberation libasound2t64 libatk-bridge2.0-0 libatk1.0-0 libatspi2.0-0 libcups2 libgbm1 libgtk-3-0 libvulkan1 libxcomposite1 libxdamage1 libxfixes3 libxkbcommon0 libxrandr2 xdg-utils

Wait until the installation is completed.

Step 8: Open a New Terminal

Keep the jwebserver terminal running.

Open a **new terminal** in Codespaces.

Navigate to the project directory if required:

cd /workspaces/STQA-Practical-12-CheckboxValidation

Step 9: Run the Selenium Test

Execute:

mvn test

The Selenium test will:

1. Identify all checkboxes.
2. Count total checkboxes.
3. Count checked checkboxes.
4. Count unchecked checkboxes.
5. Select the **Python** checkbox.
6. Deselect the **Java** checkbox.
7. Recalculate the checkbox counts.
8. Validate all results using assertions.

Step 10: Verify the Output

Expected output includes:

CHECKBOX IDENTIFICATION AND VALIDATION

Total checkboxes : 6
Checked checkboxes : 2
Unchecked checkboxes: 4

Python checkbox selected: true

Java checkbox selected after deselection: false

FINAL CHECKBOX STATUS

Checked checkboxes : 2
Unchecked checkboxes: 4

All checkbox validations passed.

At the end, you should see:

Tests run: 1, Failures: 0, Errors: 0

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BUILD SUCCESS

Result

The checkbox identification and validation was successfully performed using **Selenium**

```
====================================
CHECKBOX IDENTIFICATION AND VALIDATION
====================================
Total checkboxes   : 6
Checked checkboxes : 2
Unchecked checkboxes: 4
Python checkbox selected: true
Java checkbox selected after deselection: false
------------------------------------
FINAL CHECKBOX STATUS
Checked checkboxes : 2
Unchecked checkboxes: 4
------------------------------------
All checkbox validations passed.
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.335 s -- in practical.CheckboxValidationTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.124 s
[INFO] Finished at: 2026-10-09T07:55:21Z
[INFO] ------------------------------------------------------------------------
@dhruvsoni-android ➜/workspaces/STQA-Practical-12-CheckboxValidation (main) $
```

Checkbox Identification and Validation

Select your preferred subjects:

- [x] Java
- [ ] Python
- [x] Software Testing
- [ ] Database Management
- [ ] Web Development
- [ ] Computer Networks

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Practical 13

Dynamic Web Table Handling and Sorting Validation

Aim

To automate a dynamic web table using Selenium WebDriver and:

- Extract table data.
- Validate ascending and descending sorting.
- Filter the table and verify the filtered results.

Requirements

- GitHub Codespaces
- Java
- Maven
- Selenium WebDriver
- Chrome

Step 1: Open the Practical Repository

https://github.com/Conceicaodsouza/STQA-Practical-13-DynamicWebTable

Open the repository in **GitHub Codespaces**.

Click Code >> Codespaces >> Create codespace on new

Step 2: Check Java

Run: java -version

Java should be available.

Step 3: Check Maven

Run: mvn -version

Maven should be available.

Step 4: Check Chrome

4.1 Check whether Selenium has downloaded Chrome

Run:

find ~/.cache/selenium/chrome -type f -name chrome 2>/dev/null

You should see a path similar to:

/home/codespace/.cache/selenium/chrome/linux64/155.0.8059.39/chrome

Note: The Chrome version may be different for different students.

4.2 If Chrome is not displayed

Run:

mvn test

Selenium Manager will automatically download the required Chrome browser.

If the test fails with:

SessionNotCreatedException
Chrome instance exited

run Step 4.3 below.

4.3 Check Chrome for missing Linux libraries

First find the Chrome executable again:

find ~/.cache/selenium/chrome -type f -name chrome 2>/dev/null

Then check its required libraries.

For the current Chrome version, run:

ldd ~/.cache/selenium/chrome/linux64/155.0.8059.39/chrome | grep "not found"

If libraries are missing, you will see entries such as:

libatk-1.0.so.0 => not found
libcups.so.2 => not found
libasound.so.2 => not found

4.4 Install the required Chrome libraries

Run the following **single command**:

```
sudo apt-get update && sudo apt-get install -y \
libatk1.0-0t64 \
libatk-bridge2.0-0t64 \
libcups2t64 \
libxkbcommon0 \
libasound2t64 \
libgbm1 \
libxcomposite1 \
libxdamage1 \
libxfixes3 \
libxrandr2 \
libatspi2.0-0
```

This installs the Linux libraries required by Chrome in the Codespace.

Wait until the installation finishes and the terminal prompt returns.

4.5 Check Chrome directly

Run:

```
~/.cache/selenium/chrome/linux64/155.0.8059.39/chrome --headless=new --no-sandbox --disable-dev-shm-usage --disable-gpu --version
```

Expected result:

Google Chrome for Testing 155.0.8059.39

The version number may be different if Selenium downloaded a newer Chrome version.

Important

If the Chrome path shows a **different version**, use that version in the command.

For example, if Step 4.1 shows:

~/.cache/selenium/chrome/linux64/156.0.xxxxx.xx/chrome

then use:

~/.cache/selenium/chrome/linux64/156.0.xxxxx.xx/chrome --headless=new --no-sandbox --disable-dev-shm-usage --disable-gpu --version

---

4.6 Final Chrome dependency check

Run:

ldd ~/.cache/selenium/chrome/linux64/155.0.8059.39/chrome | grep "not found"

If **nothing is displayed**, all required libraries are available.

Chrome is now ready for Selenium.

Step 5: Start the Web Server

To allow the student to **view and interact with the webpage**, run:

jwebserver -b 0.0.0.0 -p 8000 -d src/test/resources

Keep this terminal running.

Step 6: Forward Port 8000

In GitHub Codespaces:

1. Open the **PORTS** tab.
2. Select **Forward a Port**.
3. Enter:

8000

4. Open the forwarded URL.

The **Dynamic Web Table** webpage should appear.

Webpage activities

The webpage contains:

- 10 student records.
- Student ID, Name, Course, Marks and Result.
- **Sort by Marks** button.
- **Result Filter** dropdown.

Try:

1. View the complete table.
2. Click **Sort by Marks**.
3. Click it again to reverse the sorting.
4. Select **Pass** from the filter.

Step 7: Run the Selenium Test

Open a **new terminal** so that the web server continues running.

Run:

mvn test

The Selenium program will automatically:

1. Open the dynamic web table.
2. Wait for the table rows to load.
3. Extract all 10 records.
4. Validate ascending marks.
5. Validate descending marks.
6. Filter the table for **Pass**.
7. Verify that 6 Pass records are displayed.

Expected Output

The output should contain:

DYNAMIC WEB TABLE DATA

Total rows: 10

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Ascending marks:
[29, 38, 45, 54, 61, 67, 73, 78, 85, 92]

Descending marks:
[92, 85, 78, 73, 67, 61, 54, 45, 38, 29]

FILTERED RESULTS - PASS

Filtered rows: 6

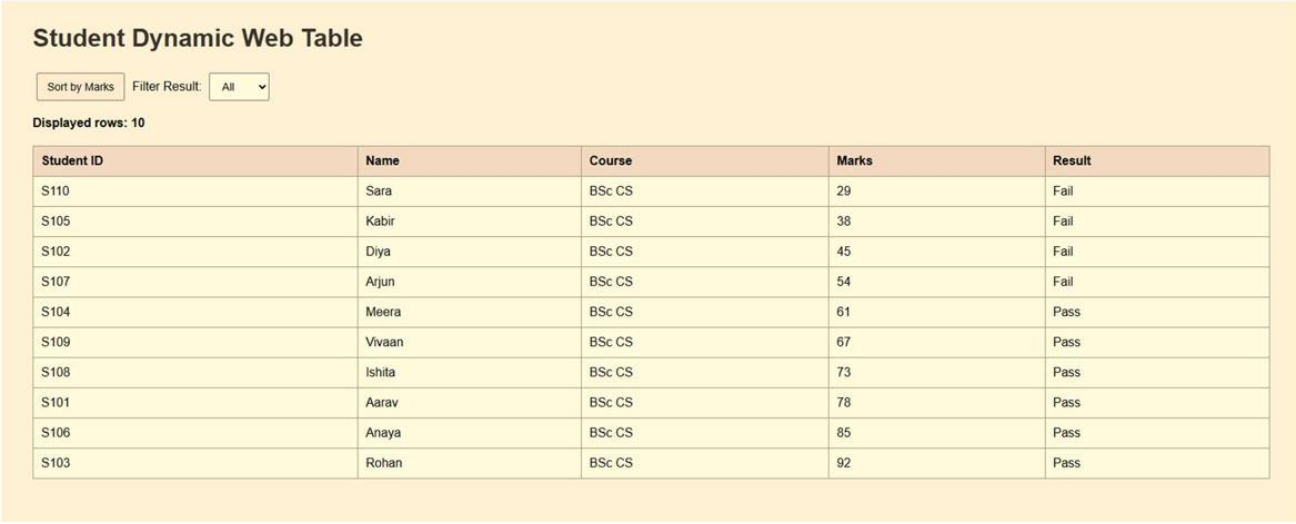
ALL VALIDATIONS PASSED

Finally:

BUILD SUCCESS

Result

The dynamic web table was successfully automated using Selenium WebDriver. Table data extraction, sorting validation, and filtered-result verification were successfully performed.



Student Dynamic Web Table

Sort by Marks Filter Result: All

Displayed rows: 10

Student ID	Name	Course	Marks	Result
S110	Sara	BSc CS	29	Fail
S105	Kabir	BSc CS	38	Fail
S102	Diya	BSc CS	45	Fail
S107	Arjun	BSc CS	54	Fail
S104	Meera	BSc CS	61	Pass
S109	Vivaan	BSc CS	67	Pass
S108	Ishita	BSc CS	73	Pass
S101	Aarav	BSc CS	78	Pass
S106	Anaya	BSc CS	85	Pass
S103	Rohan	BSc CS	92	Pass

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Student Dynamic Web Table

Sort by Marks Filter Result: Pass

Displayed rows: 6

Student ID	Name	Course	Marks	Result
S101	Aarav	BSc CS	78	Pass
S103	Rohan	BSc CS	92	Pass
S104	Meera	BSc CS	61	Pass
S106	Anaya	BSc CS	85	Pass
S108	Ishita	BSc CS	73	Pass
S109	Vivaan	BSc CS	67	Pass

Student Dynamic Web Table

Sort by Marks Filter Result: Fail

Displayed rows: 4

Student ID	Name	Course	Marks	Result
S102	Diya	BSc CS	45	Fail
S105	Kabir	BSc CS	38	Fail
S107	Arjun	BSc CS	54	Fail
S110	Sara	BSc CS	29	Fail

```
====================================
ALL VALIDATIONS PASSED
====================================
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 8.950 s -- in practical.DynamicWebTableTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  22.953 s
[INFO] Finished at: 2026-10-09T08:21:00Z
[INFO] ------------------------------------------------------------------------
@Dy1907 ➜/workspaces/STQA-Practical-13-DynamicWebTable (main) $
```